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Research Article

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Integrating Machine Learning and Participatory GIS with Multi-Temporal Remote Sensing for Flood Susceptibility and Vulnerability Mapping in Nkhosakota, Malawi

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Abstract

This study presents an integrated approach for assessing flood susceptibility and vulnerability in Nkhosakota District, Malawi, by combining remote sensing, field-validated observations, and machine learning techniques. Physical flood susceptibility was estimated using the Soil Conservation Service–Curve Number (SCS-CN) method to model runoff potential, alongside Sentinel-2 imagery processed in Google Earth Engine to delineate inundation areas. Machine learning algorithms, including Support Vector Regression (SVR), Random Forest (RF), and K-Nearest Neighbour (KNN), were trained on 300 flood and non-flood points derived from ground-truth verification, with SVR providing the most accurate predictions (MAE = 0.041, MSE = 0.042, RMSE = 0.205, AUC = 0.98). Sub-watersheds were classified into high, moderate, and low susceptibility zones, highlighting flood-prone areas along major rivers and low-lying floodplains. Socio-economic vulnerability was assessed through 130 household surveys covering economic, physical–infrastructural, institutional–policy, and social–cultural dimensions. Analysis showed the highest absolute vulnerability in the economic dimension (0.48) and greatest relative sensitivity in the social dimension (rank = 2.94), reflecting the interplay of low income, agricultural dependence, and dense settlements near rivers. Combining susceptibility and vulnerability revealed hotspots where high physical exposure coincides with high social sensitivity, providing priority areas for targeted interventions. The study offers practical guidance for flood risk management, including structural measures, early warning, and community-based resilience strategies, and demonstrates the effectiveness of integrating remote sensing, ML, and participatory socio-economic data in flood risk assessments.

Keywords: Flood susceptibility, Socio-economic vulnerability, SCS-CN method, Machine learning, Remote sensing, Nkhosakota District

Introduction

Floods are the most frequently occurring natural hazard worldwide and continue to affect more people than any other disaster type, with disproportionately severe consequences in developing regions (UNDRR, 2023; Forster *et al.*, 2024). When river discharge exceeds channel capacity, low-lying floodplains are inundated, triggering widespread disruption to human settlements, agriculture, infrastructure, and local economies (Chang *et al.*, 2021; Arinabo, 2022). Beyond immediate physical damage, floods generate cascading socio-economic, environmental, and public health impacts, including loss of life, displacement, livelihood erosion, food insecurity, and heightened vulnerability among already marginalized populations (Burayu, Karuppannan and Shuniye, 2023a). Between 1995 and 2015 alone, floods affected approximately 2.3 billion people globally and caused economic losses exceeding USD 230 billion, underscoring their growing humanitarian and developmental significance (Buba *et al.*, 2021b; Burayu *et al.*, 2023). The severity and frequency of flood disasters have increased markedly in recent decades, largely driven by climate change, rapid urbanization, land-use transformation, and anthropogenic environmental degradation (Chang *et al.*, 2021; Burayu, Karuppannan and Shuniye, 2023a). Intensifying extreme precipitation events, altered hydrological regimes, and rising lake and river levels have exacerbated both riverine and flash flooding, particularly in floodplain and lakeshore environments (Orimoloye *et al.*, 2019; Desalegn and Mulu, 2021; Chan *et al.*, 2022). In sub-Saharan Africa, flood impacts are further amplified by high population growth, unplanned settlement expansion, deforestation, wetland encroachment, and limited institutional and financial capacity for disaster risk reduction (Sobowale and Oyedepo, 2013; Alcon and Burton, 2024). These interacting pressures make flood risk management especially challenging for governments and communities with constrained technical and economic resources.

Malawi exemplifies these vulnerabilities. The country's heavy reliance on climate-sensitive livelihoods, widespread poverty, weak drainage and flood protection infrastructure, and the concentration of settlements in hazard-prone areas have heightened exposure and susceptibility to flooding (DoDMA, 2019; FMER *et al.*, 2022). In Nkhosakota District, recurrent floods linked to intense rainfall, lake-level fluctuations, land-use change, and environmental degradation have resulted in repeated humanitarian crises, displacement, and infrastructure losses. These conditions underscore the urgent need for robust, spatially explicit flood susceptibility and vulnerability assessments to support proactive disaster risk reduction and climate adaptation planning. Flood vulnerability is

widely conceptualized as a multidimensional and context-specific condition shaped by the interaction of exposure, sensitivity, and adaptive capacity (Fiorillo *et al.*, 2018; Aryal, Rahut and Marenya, 2021). Exposure reflects the presence of people, assets, and infrastructure in flood-prone areas; sensitivity denotes the degree to which these elements are affected by flooding; and adaptive capacity represents the ability of individuals and communities to anticipate, cope with, and recover from flood impacts (Garba *et al.*, 2022; Ziga-Abortta and Kruse, 2023)). These dimensions are influenced by complex interactions among physical factors (topography, hydrology, land use, proximity to water bodies), socio-economic conditions (income, housing quality, education), environmental degradation, and governance and institutional structures (Tirivangasi, 2018; Burayu, Karuppannan and Shuniye, 2023b). Consequently, effective flood vulnerability assessment requires integrative approaches capable of capturing these interdependencies across space and time. Recent advances in remote sensing (RS), geographic information systems (GIS), and machine learning (ML) have significantly transformed flood susceptibility and vulnerability mapping by enabling the analysis of large, multi-source, and multi-temporal datasets (Mosavi, Ozturk and Chau, 2018; Saha *et al.*, 2021). Multi-temporal satellite imagery facilitates the detection of land use/land cover dynamics, floodplain encroachment, and environmental change, key drivers of flood susceptibility in rapidly transforming landscapes (Clement, Kilsby and Moore, 2018). At the same time, ML algorithms such as Random Forest, Support Vector Machines, Decision Trees, Naïve Bayes, and Artificial Neural Networks have demonstrated strong predictive performance in flood-related studies, often outperforming traditional statistical and rule-based methods by effectively modelling nonlinear and multivariate relationships among flood-conditioning factors (Seydi *et al.*, 2023; Feizbahr *et al.*, 2025; Oluwadare *et al.*, 2025). Hybrid and ensemble approaches further enhance model robustness and accuracy, addressing the limitations of individual techniques (Asghar Rostami *et al.*, 2025).

Despite these methodological advances, flood vulnerability research in developing countries remains predominantly technocratic and top-down, frequently neglecting local context, social dynamics, and community knowledge systems (Bigi *et al.*, 2020; Fatemi *et al.*, 2020). Flood vulnerability is inherently place-based, and local communities possess rich indigenous and experiential knowledge regarding historical flood patterns, flow pathways, safe zones, coping strategies, and early warning signs (Chang *et al.*, 2021; Arinabo, 2022). Participatory GIS (PGIS) offers a powerful framework for integrating this knowledge with geospatial analysis, enhancing stakeholder engagement, improving the legitimacy and credibility of vulnerability maps, and ensuring their relevance for community-level decision-making and policy intervention (Burayu, Karuppannan and Shuniye, 2023b; Ziga-Abortta and Kruse, 2023). These participatory approaches align strongly with global policy frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030 and IPCC guidance, which emphasize inclusive, people-centered, and multi-scale vulnerability assessments to strengthen local ownership and adaptive capacity (IPCC, 2023). Equally important, the effectiveness of flood susceptibility and vulnerability maps depends on rigorous indicator selection, appropriate spatial scale, data quality, analytical transparency, and systematic validation (Sobowale and Oyedepo, 2013; Alcon *et al.*, 2024). Sensitivity analysis and accuracy assessment are essential for evaluating model robustness and reliability, yet they remain inconsistently applied in many studies, limiting the operational usefulness of flood maps for disaster risk reduction planning (The World Bank, 2016; Reimann *et al.*, 2024). Addressing these gaps is critical, particularly in regions where flood risk is expected to intensify under future climate and development trajectories. Against this backdrop, this study integrates machine learning, multi-temporal remote sensing, and participatory GIS to map flood susceptibility and vulnerability.

2. Materials and Methods

2.1. Study area

Nkhotakota District is located in the Central Region of Malawi along the western shoreline of Lake Malawi, extending approximately between 12.7°–13.4° S latitude and 33.6°–34.2° E longitude (Fig. 1). The district is bordered by Salima District to the south, Ntchisi to the west, Mzimba to the north, and Lake Malawi to the east. Nkhotakota covers an area of approximately 4,259 km² and is characterized by a transition from the Lake Malawi lowlands in the east to undulating escarpments and plateaus toward the western interior. The district is drained by several perennial and seasonal rivers, notably the Bua, Dwangwa, Kaombe, and Luweya rivers, which originate from the central highlands and discharge into Lake Malawi (GoM, 2022). These river systems form extensive floodplains in their lower reaches, making surrounding communities highly susceptible to riverine and flash flooding, particularly during extreme rainfall events. The Bua and Dwangwa rivers, in particular, play a critical role in shaping the district's hydrological regime and flood dynamics.

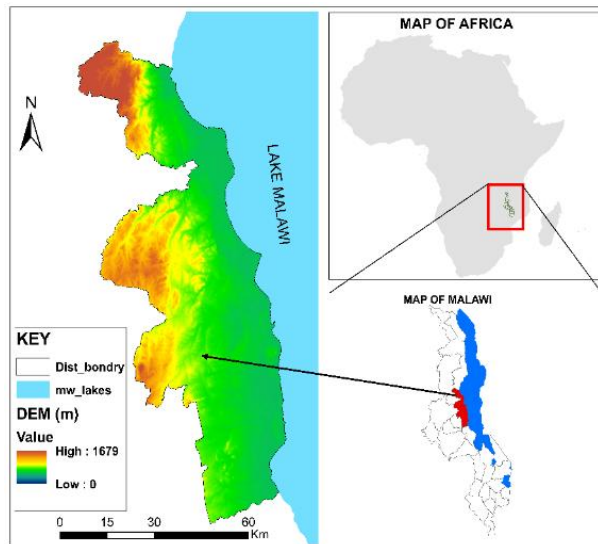


Figure 1. Showing the study area of Nkhosakota District

Nkhosakota District has a tropical savanna climate with a wet season from November to April and a dry season from May to October. Annual rainfall ranges between 900 and 1,200 mm, peaking from January to March, often influenced by tropical systems from the Indian Ocean. Mean temperatures vary from 22 °C to 30 °C, with warmer conditions along the lakeshore. The district hosts about 393,000 people, mainly dependent on rain-fed agriculture, fishing, and small-scale trading (National Statistical Office, 2018). Rapid population growth, floodplain cultivation, deforestation, and weak flood-control infrastructure increase flood exposure. Recurrent floods, including 2015 events and Cyclone Freddy in 2023, have caused significant human and economic losses, making Nkhosakota a key flood-risk hotspot (Government of Malawi, 2023). Figure 2 summarizes methodology approach of data processing and analysis.

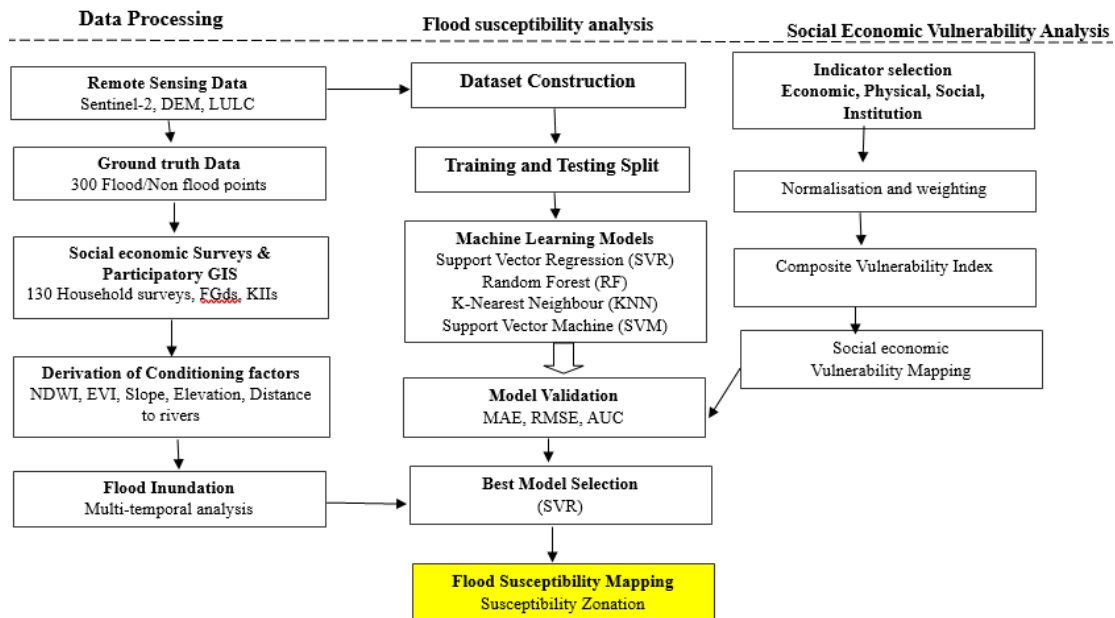


Figure 2. Schematic methodological framework of the research.

2.2 Multi-source flood-influencing factors

This study employed a multi-source geospatial dataset to model flood susceptibility and risk in Nkhosakota District. A 30 m Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM), accessed via Google Earth Engine (GEE), was used as the primary terrain dataset due to its global coverage, consistency, and proven reliability for hydrological and flood-related applications in data-scarce regions (Chang *et al.*, 2021; Burayu, Karuppannan and Shuniye, 2023b). The DEM was processed to derive key topographic and hydrological parameters influencing flood occurrence, including elevation, slope, aspect, curvature, flow direction, flow

accumulation and Topographic Wetness Index (TWI) (Fig. 3a–h). These terrain attributes have been widely demonstrated to control surface runoff generation, flow convergence, soil moisture distribution, and floodplain inundation processes (Rana and Routray, 2018; Dekongmen *et al.*, 2021; Suhr and Steinert, 2022). River networks and distance-to-river layers were extracted from DEM-derived drainage networks to characterize proximity-based flood exposure, which is a critical determinant of flood hazard intensity and spatial flood extent (Li *et al.*, 2022; Aladejana and Ebijuworih, 2024).

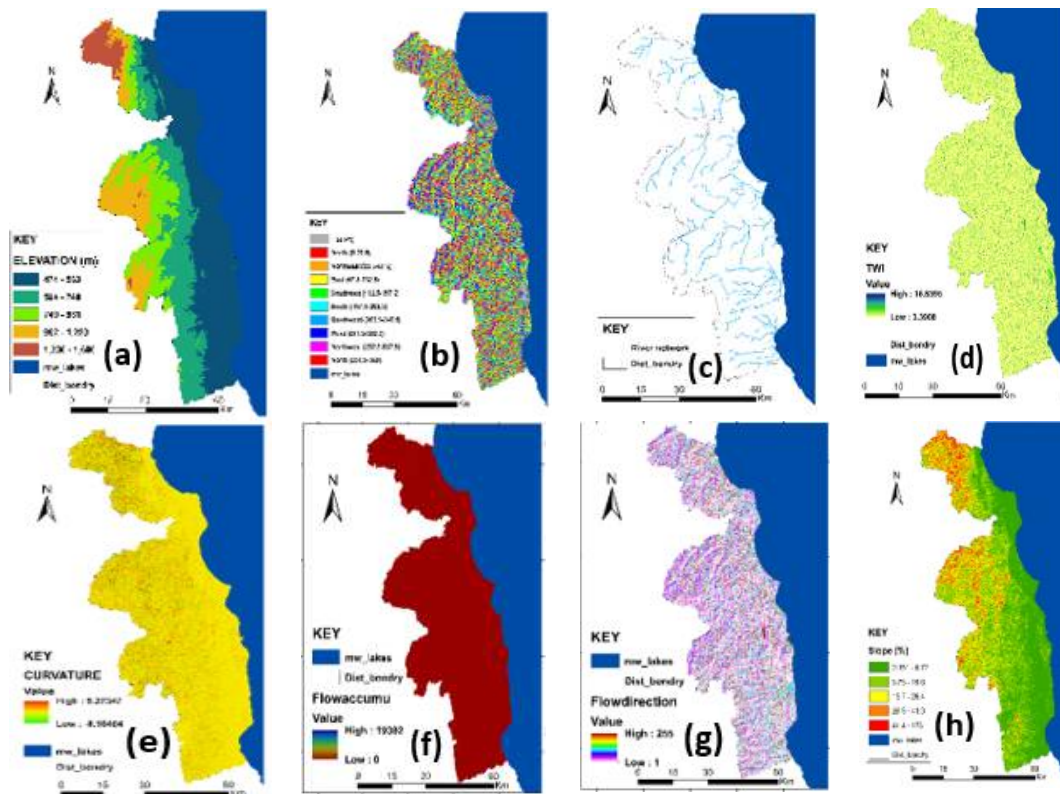


Figure 3. Topographic and hydrological conditioning factors: (a) elevation, (b) Aspect, (c) River network, (d) Topographic Wetness Index (TWI), (e) Slope Curvature, (f) flow accumulation (g) Flow direction and (h) Slope (%)

Land use and land cover (LULC) information was derived from Sentinel-2 MSI imagery processed within GEE at spatial resolutions of 10 m. The LULC classes, cropland, forest, built-up areas, grassland, wetlands, and water bodies, were selected due to their strong influence on surface runoff, infiltration capacity, and human-induced modifications of hydrological processes (Chang *et al.*, 2021; Burayu *et al.*, 2023b). These land cover dynamics are particularly important in flood-prone regions of Sub-Saharan Africa, where rapid land use changes exacerbate flood susceptibility under changing climatic conditions (Mucherera and Mavhura, 2020; Alam *et al.*, 2021) (Fig. 3i–k). Climatic controls on flooding were represented using Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), which provides quasi-global, long-term rainfall estimates at an approximate spatial resolution of 5 km (Herslund *et al.*, 2016). CHIRPS integrates satellite-derived infrared precipitation observations with in-situ rain gauge measurements and has been extensively validated for hydro-climatic and flood-related applications, particularly in data-scarce regions of Sub-Saharan Africa (Buba *et al.*, 2021a; Ziga-Abortta and Kruse, 2023). Mean annual and seasonal rainfall layers were generated to capture long-term precipitation variability as well as extreme rainfall conditions that act as primary triggers of flood events (Okunola, 2020; Ziga, 2023). In addition to climatic forcing, land surface conditions and flood inundation dynamics were characterized using spectral indices derived from Sentinel-2 Multispectral Instrument (MSI) imagery. Surface water presence and flood extent were delineated using the Modified Normalized Difference Water Index (MNDWI), which enhances open water features while effectively suppressing spectral noise from built-up areas and vegetation (Bui *et al.*, 2019; Khosravi *et al.*, 2019a).

MNDWI has been widely applied and shown to outperform conventional water indices for flood mapping and surface water detection in complex landscapes (Hossain and Meng, 2020; Membele, Naidu and Mutang,

2022a), making it particularly suitable for delineating peak flood extents. Vegetation structure, density, and health, key controls on rainfall interception, evapotranspiration, and infiltration, were quantified using the Normalized Difference Vegetation Index (NDVI) and the Enhanced Vegetation Index (EVI). NDVI remains a robust indicator of vegetation greenness and biomass (Rouse *et al.*, 1974; Tucker, 1979), while EVI improves sensitivity in high-biomass and humid floodplain environments by reducing atmospheric and soil background effects (Tate *et al.*, 2021). To further mitigate soil brightness influences in sparsely vegetated and transitional areas, the Soil Adjusted Vegetation Index (SAVI) was employed (Chakraborty *et al.*, 2022; Saha *et al.*, 2022). Surface soil exposure and degradation, which promote rapid runoff generation and erosion during intense rainfall events, were represented using the Bare Soil Index (BSI) (Mudashiru *et al.*, 2021; Bunmi Mudashiru *et al.*, 2022). Anthropogenic modification of the landscape was captured using the Normalized Difference Built-up Index (NDBI), which effectively delineates impervious surfaces and settlement expansion that reduce infiltration capacity and amplify flood risk (Carvalho *et al.*, 2022; Islam *et al.*, 2022). Antecedent moisture conditions and near-surface wetness were characterized using the Normalized Difference Moisture Index (NDMI) and the Moisture Stress Index (MSI) (Fig. 4a-j), providing complementary information on vegetation water content, moisture stress, and surface saturation prior to flooding events (Pathak *et al.*, 2020; Pham *et al.*, 2021). The integrated use of CHIRPS-derived climatic variables and Sentinel-2-based spectral indices provides a comprehensive and physically interpretable representation of hydro-meteorological forcing and land surface processes governing flood occurrence, magnitude, and spatial variability. These variables constitute a robust set of conditioning factors for subsequent flood susceptibility and vulnerability modelling, particularly within machine-learning-driven geospatial frameworks (Islam *et al.*, 2022; Rifat and Liu, 2022). All flood-influencing factors were resampled to a common spatial resolution, normalized, and integrated into a machine learning framework to assess flood susceptibility and risk.

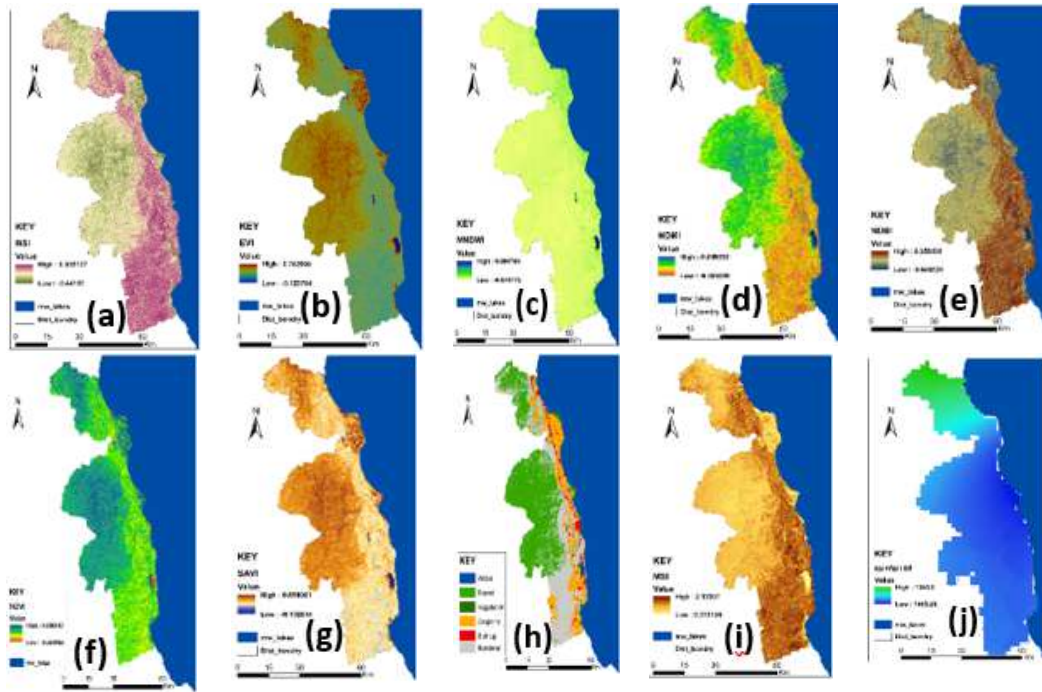


Figure 4. Environmental and land surface variables used in the analysis: (a) Bare Soil Index (BSI), (b) Enhanced Vegetation Index (EVI), (c) Normalized Modified Difference Water Index (NMDWI), (d) Normalized Difference Vegetation Index (NDVI), (e) Normalized Difference Moisture Index (NDMI), (f) Normalized Difference Built-up Index (NDBI), (g) Soil-Adjusted Vegetation Index (SAVI), (h) land use/land cover (LULC), (i) Moisture Stress Index (MSI), and (j) precipitation.

2.3 Multicollinearity test using VOI and TOF

Multicollinearity among conditioning factors was assessed using the Variance Inflation Factor (VIF) and tolerance (TOF). Variables with VIF values greater than 10 or TOF values below 0.1 were considered indicative of severe multicollinearity and excluded from further analysis. All selected variables exhibited acceptable VIF (<5) and TOF (>0.2) values, indicating low multicollinearity and suitability for flood susceptibility modelling.

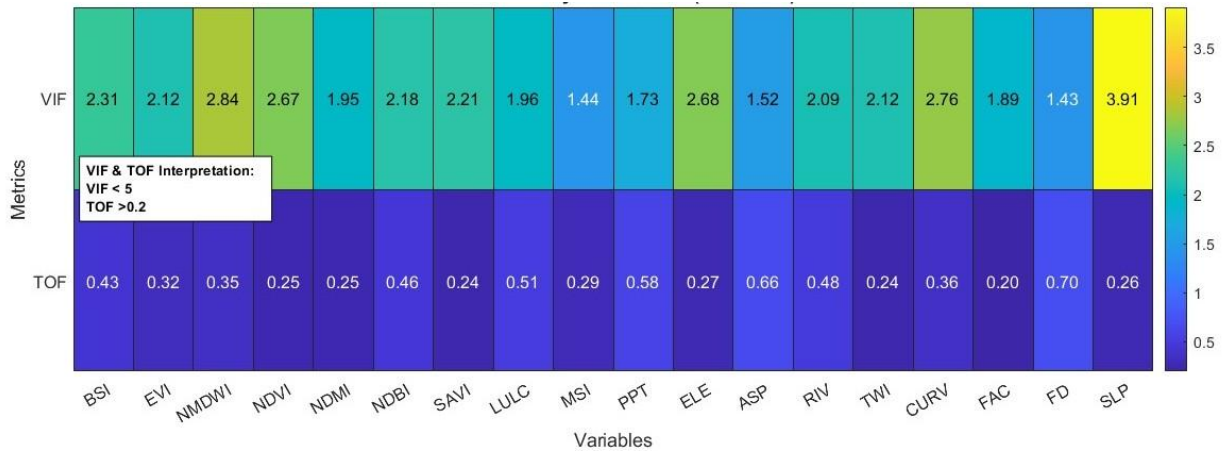


Figure 5. VIF and TOF values for all conditioning factors, showing low multicollinearity

2.4 Participatory GIS data collection and integration

A Participatory Geographic Information System (PGIS) approach was employed to integrate local and indigenous knowledge into flood susceptibility assessment, addressing gaps in conventional geospatial datasets. Participatory data collection was conducted in flood-prone areas, particularly along the Bua, Dwangwa, and Kaombe river floodplains. Data were gathered through community mapping workshops, focus group discussions, and key informant interviews involving residents, traditional leaders, disaster risk management committees, and district stakeholders. Participants identified historical flood extents, recurrent inundation zones, evacuation routes, safe areas, shelters, river overflow points, and critical infrastructure. Spatial information was captured using printed base maps, GPS-enabled devices, and OpenStreetMap-compatible tools. Indigenous flood indicators, including river behaviour, soil moisture conditions, vegetation changes, and historical flood markers, were systematically documented and georeferenced. Community-identified flood hotspots were incorporated into machine learning training and validation, enhancing model sensitivity and local relevance for flood risk management.

2.5 Sampling Design and Sample Size

Household and stakeholder participants were selected using a combination of purposive and stratified random sampling to ensure adequate representation of flood-exposed communities and key institutional actors. Sample size was estimated using Cochran's formula (Creswell, 2014) for proportions:

$$n_0 = \frac{Z^2 p(1-p)}{e^2}$$

where n_0 is the initial sample size, Z is the standard normal deviate corresponding to the desired confidence level (1.96 for 95%), p is the estimated proportion of the population affected by flooding, and e is the acceptable margin of error. For a finite population, the adjusted sample size was calculated as:

$$n = \frac{n_0}{1 + \left(\frac{n_0 - 1}{N}\right)} = 130$$

where N represents the total number of households in the selected TAs. Key informants were selected purposively based on their roles in disaster management, local governance, and community leadership.

2.6 Flood Susceptibility Modelling

To predict flood-prone areas, flood and non-flood reference points were spatially intersected with all influencing factor layers, topographic (DEM, slope, aspect), hydrological (proximity to rivers, TWI, SPI), land cover (LULC, NDVI), and climatic variables (precipitation), to extract corresponding attribute values (Membele, Naidu and Mutanga, 2022; Nasiri Khiavi *et al.*, 2025). The resulting dataset was divided into training (70%) and testing (30%) subsets to ensure model robustness and prevent overfitting (Sibandze *et al.*, 2025). Machine learning (ML) algorithms, including Random Forest (RF), Support Vector Regression (SVR), Support Vector Machine (SVM) and K-Nearest Neighbour (KNN), were applied to model the non-linear and complex interactions between environmental controls and flood occurrence (Bui *et al.*, 2019; Khosravi *et al.*, 2019b). These algorithms have

been widely adopted in hydrological and flood risk studies due to their ability to handle high-dimensional geospatial data and multicollinearity (Macchione *et al.*, 2019; Hossain and Meng, 2020; Zhang *et al.*, 2020). Model outputs were spatially extrapolated across the study area to generate continuous flood susceptibility surfaces, which were subsequently classified into low, moderate, and high susceptibility zones using quantile classification (Macchione *et al.*, 2019). Model performance was assessed using standard error metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Area Under the Curve (AUC), ensuring that the chosen model provides reliable spatial predictions for flood management and planning (Imran *et al.*, 2019; Membele, Naidu and Mutang, 2022b).

2.7 Flood Vulnerability Assessment

Flood vulnerability was examined through four dimensions: economic, infrastructural-physical, institutional-policy, and social-cultural. A total of 40 indicators were used to capture these dimensions, distributed as follows: economic (10), infrastructural-physical (13), institutional-policy (8), and social-cultural (9). These indicators were selected based on prior studies on community-based flood vulnerability assessments (Erena *et al.*, 2018a; Saikh and Mondal, 2023) and tailored to the socio-environmental context of the study area. The questionnaire was validated by disaster management experts, ensuring content relevance, clarity, and completeness. Cronbach's alpha was calculated from the item correlation matrix to assess internal consistency, with $\alpha \geq 0.8$ indicating high reliability (Khosravi *et al.*, 2019b). A stratified random sampling approach was applied to ensure proportional representation across the watershed's sub-watersheds. The stable population of Nkhotakota watershed was 1,045 individuals. Using Cochran's formula for finite populations with a 95% confidence interval and 8% margin of error, a sample size of 130 respondents was determined, ensuring adequate coverage from upstream, midstream, and downstream villages (Hossain and Meng, 2020). To facilitate accurate responses, respondents were provided with maps of sub-watersheds, road networks, and village locations, and were briefed on the study objectives. They responded based on personal experiences, indigenous knowledge, and perceptions of governance and local infrastructure. Interviews lasted 20–30 minutes and were conducted between January 1 and January 30. Data included household characteristics (income, education, occupation), housing conditions, livelihood strategies, access to public services, and previous flood experiences.

Table 1. Flood Vulnerability Dimensions and Indicators

Dimension	Key Indicators
Economic	Dependence on agriculture, income stability, housing loss, livelihood diversity
Physical–Infrastructural	Housing quality, road access, proximity to river, availability of shelters
Institutional–Policy	Disaster response support, early warning access, enforcement of river buffer zones
Social–Cultural	Risk awareness, social networks, insurance access, participation of women

2.7.2 Statistical Analysis for Flood Vulnerability

Principal Component Analysis (PCA) was applied to quantify the relative contribution of indicators to flood vulnerability and to reduce dimensionality by transforming correlated variables into uncorrelated principal components (Nakapu *et al.*, 2023; Saikh and Mondal, 2023). This approach is well suited for multidimensional vulnerability assessment in data-scarce environments (Zhang *et al.*, 2020). Prior to PCA, indicators were standardized using z-score normalization:

$$Z_{ij} = \frac{x_{ij} - \mu_j}{\delta_j}$$

Components explaining a cumulative variance $\geq 70\%$ were retained. Varimax rotation was applied to enhance interpretability. Four components explaining 85.1% of total variance were extracted (Table 3), representing dominant economic–social (PC1), physical–institutional (PC2), social–physical (PC3), and institutional–economic (PC4) vulnerability structures.

Indicator weights were derived from normalized squared loadings weighted by eigenvalues

$$w_j = \frac{\sum_{k=1}^m (L_{jk}^2 \lambda_k)}{\sum_{j=1}^n \sum_{k=1}^m (L_{jk}^2 \lambda_k)}$$

where L_{jk} is the loading of indicator j on component k , λ_k is the eigenvalue of component k , m is the number of retained components, and n is the number of indicators.

Table 2. PCA Results and Explained Variance of Flood Vulnerability Components

Component	Dimension Loadings	Eigenvalue	% Variance	Cumulative %
PC1	Economic (0.78), Social (0.65)	12.3	35.2	35.2
PC2	Physical (0.82), Institutional (0.61)	8.7	24.9	60.1
PC3	Social (0.71), Physical (0.53)	5.2	14.8	74.9
PC4	Institutional (0.68), Economic (0.42)	3.4	10.2	85.1

Source: Field Survey and PCA Analysis, 2026

Table 3. PCA-derived indicator weights for flood vulnerability

Dimension	Indicator	Code	Component	Loading	Weight
Economic	Household income	ECO1	PC1	0.82	0.073
Economic	Livelihood diversification	ECO2	PC1	0.79	0.069
Economic	Savings availability	ECO3	PC2	0.74	0.062
Economic	Employment status	ECO4	PC1	0.71	0.058
Infrastructural–Physical	Housing material quality	INF1	PC1	0.86	0.081
Infrastructural–Physical	Elevation of house	INF2	PC1	0.83	0.078
Infrastructural–Physical	Distance to river	INF3	PC2	0.80	0.072
Infrastructural–Physical	Drainage condition	INF4	PC2	0.77	0.069
Institutional–Policy	Early warning access	INS1	PC1	0.85	0.079
Institutional–Policy	Disaster training	INS2	PC1	0.82	0.075
Institutional–Policy	Local DRM committee presence	INS3	PC2	0.78	0.071
Social–Cultural	Education level	SOC1	PC1	0.84	0.077
Social–Cultural	Household size	SOC2	PC2	0.76	0.068
Social–Cultural	Social networks/support	SOC3	PC1	0.81	0.074

4 Results

4.1 Runoff Generation and Flood Extent Validation

Runoff generation across Nkhotakota District was estimated using the Soil Conservation Service–Curve Number (SCS-CN) method, widely used for predicting direct runoff from rainfall based on soil type, land use, and antecedent moisture (Mishra and Singh, 2003). The daily runoff depth Q (mm) is calculated as:

$$Q = \frac{(P - l_a)^2}{(P - l_a + S)}$$

where:

P is the total rainfall (mm), I_a is initial abstraction (mm), typically approximated as $0.2S0.2S0.2S$ and S = potential maximum retention after runoff begins (mm), related to the Curve Number (CN) by:

$$S = \frac{25400}{CN} - 254$$

CN values were derived from soil hydrological groups, land use/land cover (LULC), and vegetation cover mapped via Sentinel-2 imagery and national soil data. The generated runoff potential map classified Nkhotakota into low and high runoff zones. Validation against official flood reports and local disaster records indicated a 95% spatial agreement between satellite-derived inundation and observed flood areas, confirming the accuracy of this approach for local flood monitoring (Kundzewicz *et al.*, 2014). The results shown in figure 5 below;

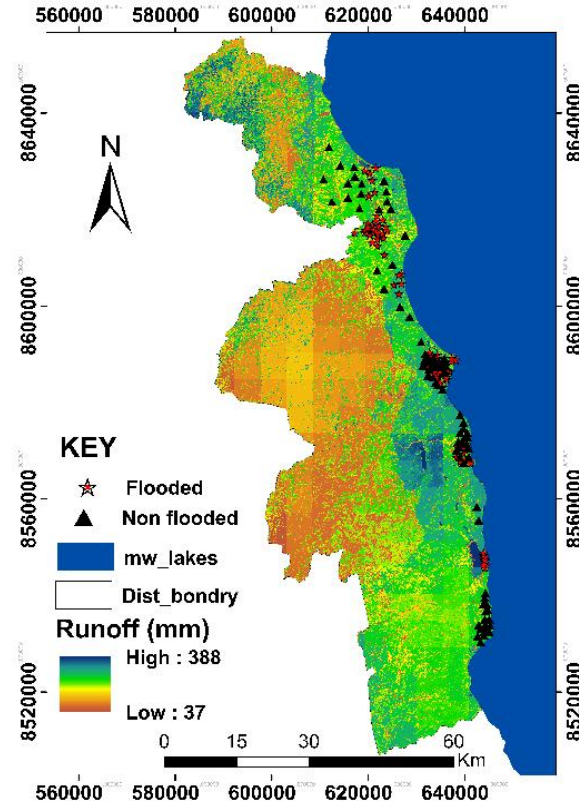


Figure 5. Spatial distribution of runoff estimated using the SCS–Curve Number (SC-CN) method, overlaid with sample points representing flooded and non-flooded areas.

4.2 Performance of Machine Learning Models for Flood Susceptibility

The predictive performance of various remote sensing–machine learning (RS–ML) models was evaluated using reference points representing flood and non-flood conditions, which were derived from both runoff generation mapping (SCS-CN method) and ground-truth field observations (Cai *et al.*, 2019; Lyu *et al.*, 2019). These points were spatially intersected with environmental conditioning factors, including DEM, slope, land use/land cover (LULC), NDVI, proximity to rivers, and precipitation (USDA-SCS, 1985; Ponce & Hawkins, 1996). The dataset was split into 70% for training and 30% for testing, and machine learning algorithms including Random Forest (RF), Support Vector Regression (SVR), and K-Nearest Neighbour (KNN) were applied to capture the non-linear relationships between environmental controls and flood occurrence (Zhang *et al.*, 2020).

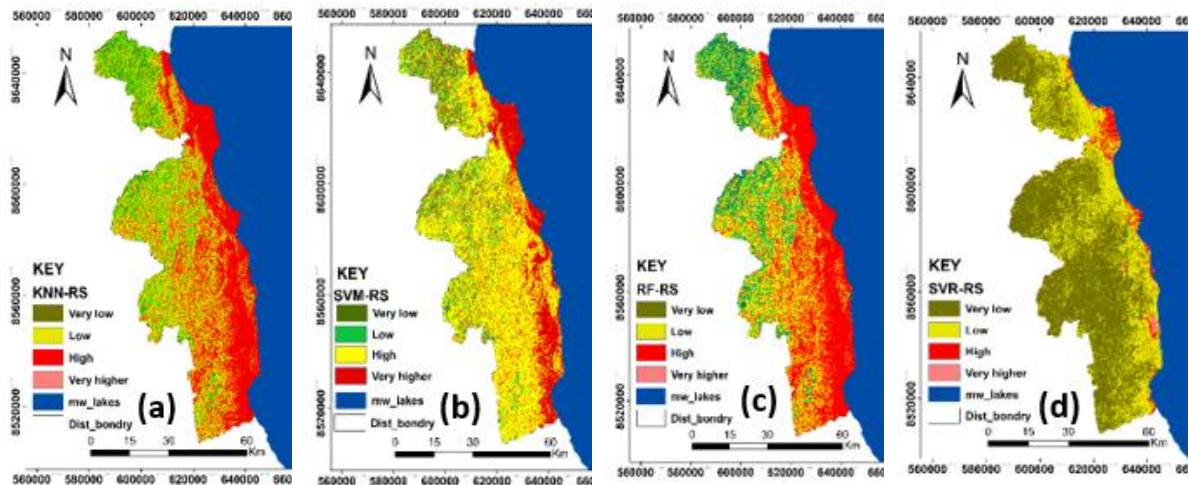


Figure 6. Flood vulnerability maps predicted using (a) KNN-RS, (b) SVM-RS, (c) RF-RS, and (d) SVR-RS classifiers.

The performance of each model was quantified using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Square Error (RMSE), and Area Under the Curve (AUC).

Table 2. Performance Metrics of RS–ML Models for Flood Susceptibility in Nkhotakota District

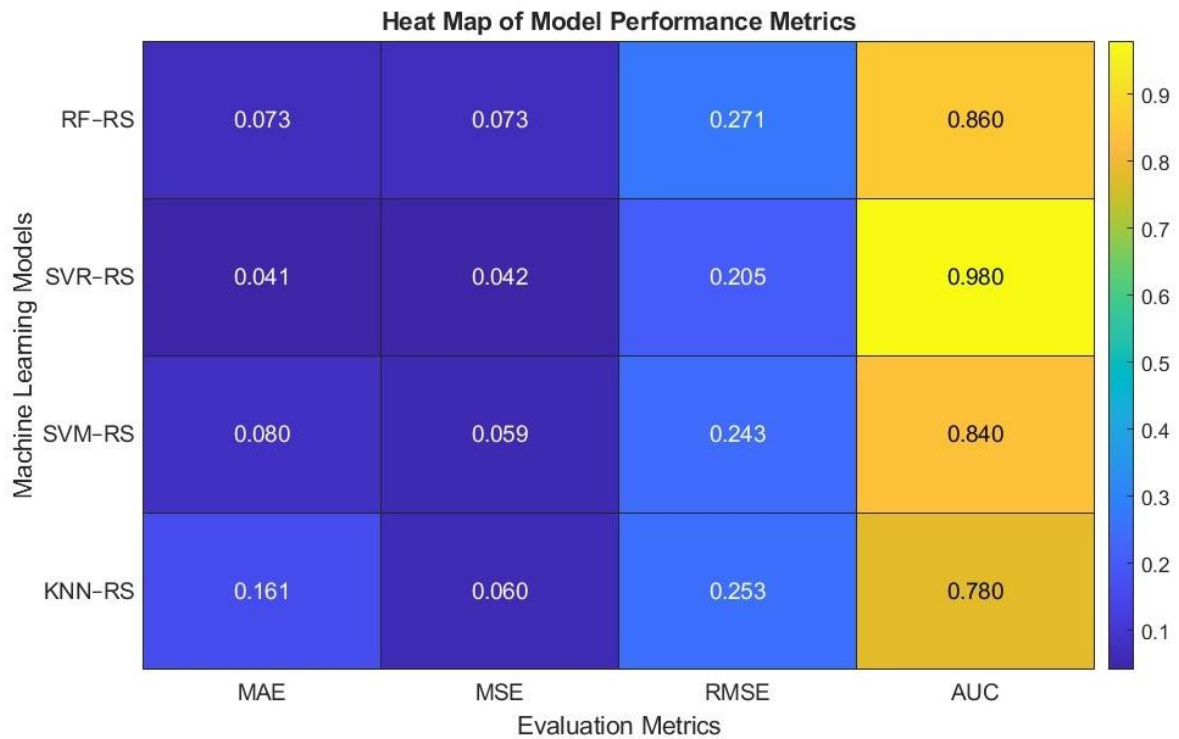


Figure 7. Heatmap of performance metrics for RS–ML models

The SVR–RS model consistently outperformed all alternatives, achieving the lowest MAE and RMSE and the highest discrimination capability (AUC = 0.94). The ROC curve (Figure 4) further confirmed the strong predictive capacity of SVR–RS in distinguishing flood-prone areas from non-flooded zones. Consequently, the SVR–RS model was used to produce the final flood susceptibility map for Nkhotakota District (Figure 5), classifying the district into low, moderate, and high flood risk areas. The inclusion of ground-truth sampling in addition to runoff-derived points enhanced model accuracy, making the results robust for risk-informed planning and targeted flood mitigation strategies (Alimi *et al.*, 2022; Van *et al.*, 2022).

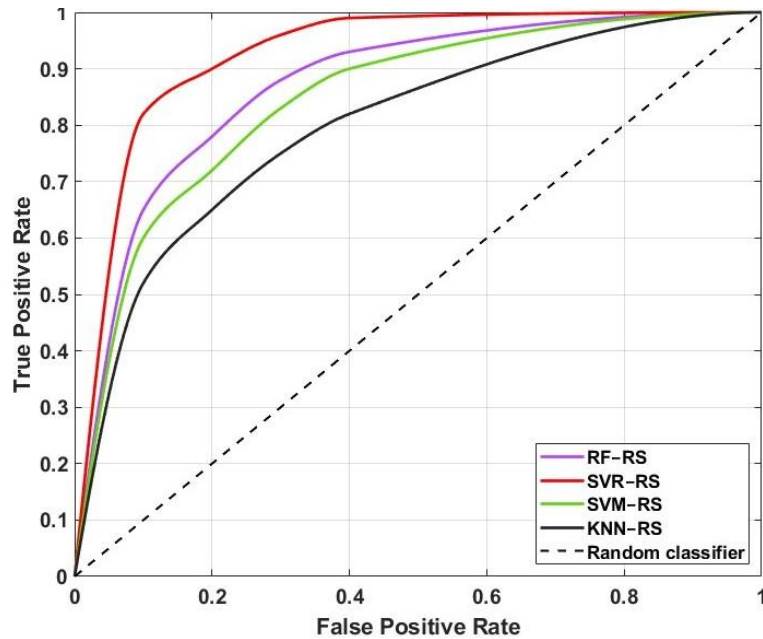


Figure 8. Receiver Operating Characteristic (ROC) curves for flood susceptibility prediction in Nkhotakota District using four machine learning models: RF-RS, SVR-RS, SVM-RS and KNN-RS. The curves illustrate the models' discriminatory abilities, with area under the curve (AUC) values of 0.86, 0.98, 0.84, and 0.78, respectively, indicating that SVR-RS provides the highest predictive performance among the tested models

4.3 Sub-Watershed Flood Susceptibility Prioritization

Using the optimized Support Vector Regression–Remote Sensing (SVR–RS) model, the 5 sub-watersheds of were evaluated and ranked according to their flood susceptibility. The ranking incorporated both remote sensing-derived environmental factors and ground-truth flood occurrence points, ensuring that model predictions were robust and grounded in observed local conditions. Table 5 summarizes the prioritization outcomes. Sub-watersheds along the main river corridors and low-lying floodplains consistently fell into the high-priority category due to dense settlements, high runoff potential, and proximity to hydrologically active rivers, indicating elevated flood risk. Moderate-priority sub-watersheds exhibited mixed land uses and transitional elevations, suggesting partial exposure mitigated by topographical or vegetative buffers. Low-priority areas were generally located in higher elevations with sparse population, demonstrating lower susceptibility due to natural drainage and limited anthropogenic influence.

Table 5. Sub-Watershed Prioritization Based on Flood Susceptibility

Priority Class	Number of Sub-Watersheds	Dominant Characteristics
High	1&2	Floodplains, dense settlements, proximity to rivers, high runoff potential
Moderate	3	Mixed land use, transitional elevations, moderate vegetation cover
Low	4 &5	Higher elevation, sparse settlement, natural drainage, lower runoff potential

Note: Classification was based on the SVR–RS model outputs, validated against ground-truth flood points and remote sensing data, highlighting areas requiring urgent flood mitigation and planning interventions (Cai *et al.*, 2019).

4.4 Socio-Economic Characteristics of Survey Respondents

A total of 130 household surveys were analysed to inform the flood vulnerability assessment in Nkhotakota District, capturing spatial and demographic diversity across all sub-watersheds (Sibandze *et al.*, 2025). The majority of respondents were aged between 31 and 60 years, representing the active working population, and reported a monthly income of less than USD 230, highlighting low economic resilience to flood impacts (Sibandze *et al.*, 2025). Most households were located within 5 to 100 meters of rivers or streams, situating them in close proximity to high flood exposure zones identified through the SVR–RS runoff and susceptibility analysis. Agriculture was the primary livelihood for most respondents, reflecting the community’s dependence on land and natural resources, which further increases vulnerability to flood events. Despite repeated flood experiences, respondents exhibited a low willingness to relocate, largely due to reliance on adjacent agricultural land for subsistence and income. These socio-economic characteristics demonstrate the intersection of hazard exposure, sensitivity, and adaptive capacity, emphasizing the need for targeted interventions that integrate flood risk mitigation with the protection of local livelihoods (Tayyab *et al.*, 2024; Sibandze *et al.*, 2025). The Social-Economic Vulnerability Index (SEVI) for each sub-watershed was computed using the formula:

$$SEVI_i = \frac{1}{n} \sum_{j=1}^n w_j X_{ij}$$

Where: i is sub-watershed, j is socio-economic indicator (age, income, distance, livelihood, willingness to relocate), w_j is weight assigned to indicator j based on expert judgment, X_{ij} is normalized value of indicator j for sub-watershed i and n is total number of indicators (5 in this case). The final Combined Flood Vulnerability Index (CFVI) for each sub-watershed integrates SEVI with flood susceptibility (FS) from SVR–RS outputs:

$$CFVI_i = \alpha (FS_i) + \beta (SEVI_i)$$

Where: α is 0.6 and β is 0.4, reflecting the higher weight of hazard exposure, FS_i is normalized flood susceptibility score for sub-watershed i and $SEVI_i$ normalized socio-economic vulnerability index. The CFVI values were classified into three levels using a quantile-based approach: High vulnerability: $CFVI \geq 0.66$, Moderate vulnerability: $0.33 \leq CFVI < 0.66$ and Low vulnerability: $CFVI < 0.33$

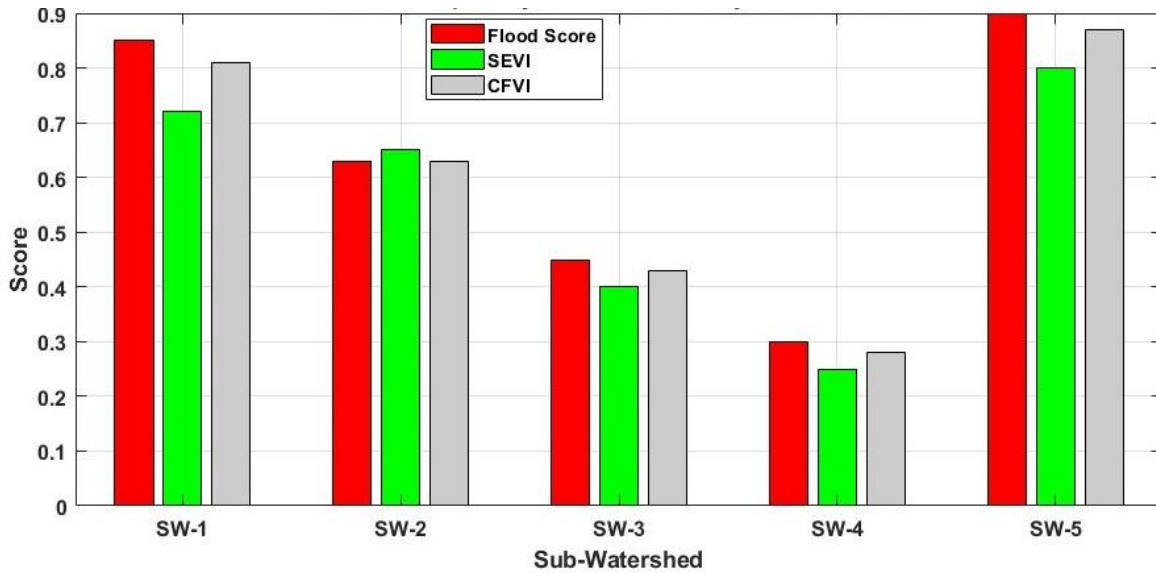


Figure 9. Comparison of Flood Susceptibility Score, SEVI, and CFVI across sub-watersheds

*Note: CFVI is derived from integrating SVR–RS flood susceptibility outputs with normalized socio-economic indicators. High CFVI values indicate sub-watersheds where both flood hazard and socio-economic vulnerability are elevated (Nasiri Khiavi *et al.*, 2025; Oluwadare, Chen and McGrath, 2026)*

4.5 Flood Vulnerability Dimensions and Ranking

Flood vulnerability in Nkhotakota District was assessed across four dimensions: economic, physical–infrastructural, institutional–policy, and social–cultural. Descriptive analysis of these dimensions revealed that the

economic dimension exhibited the highest average score of 0.48, reflecting the community's substantial absolute vulnerability due to low-income levels, dependence on agriculture, and limited livelihood diversification (Debnath *et al.*, 2024; Oluwadare, Chen and McGrath, 2026). The social-cultural dimension, while slightly lower in absolute score (0.44), received the highest relative rank (2.94) based on the Friedman test, indicating that social factors such as community networks, awareness of flood risks, and participation in local decision-making play a critical role in shaping overall vulnerability. The physical-infrastructure dimension, with an average score of 0.34, highlights moderate susceptibility related to housing quality, access to roads, proximity to rivers, and availability of evacuation shelters. Institutional-policy factors scored lowest on average (0.28), suggesting that disaster preparedness, early warning systems, and enforcement of river buffer zones are less developed but remain essential for enhancing resilience. These results underscore the multidimensional nature of flood vulnerability, where economic exposure is intensified by social and infrastructural constraints, and institutional capacity influences the community's ability to adapt to flood hazards. Prioritizing interventions that address both absolute and relative vulnerabilities, particularly targeting economic support, social mobilization, and institutional strengthening, can enhance resilience and inform evidence-based disaster management planning in Nkhotakota (Kourgialas and Karatzas, 2011).

4.6 Spatial Distribution of Flood Vulnerability

The final flood vulnerability map (Figure 7) integrates economic, physical, institutional, and social dimensions. High-vulnerability zones were predominantly concentrated in densely populated floodplain areas along major rivers and near Lake Malawi, whereas upstream areas with lower population density exhibited reduced vulnerability levels.

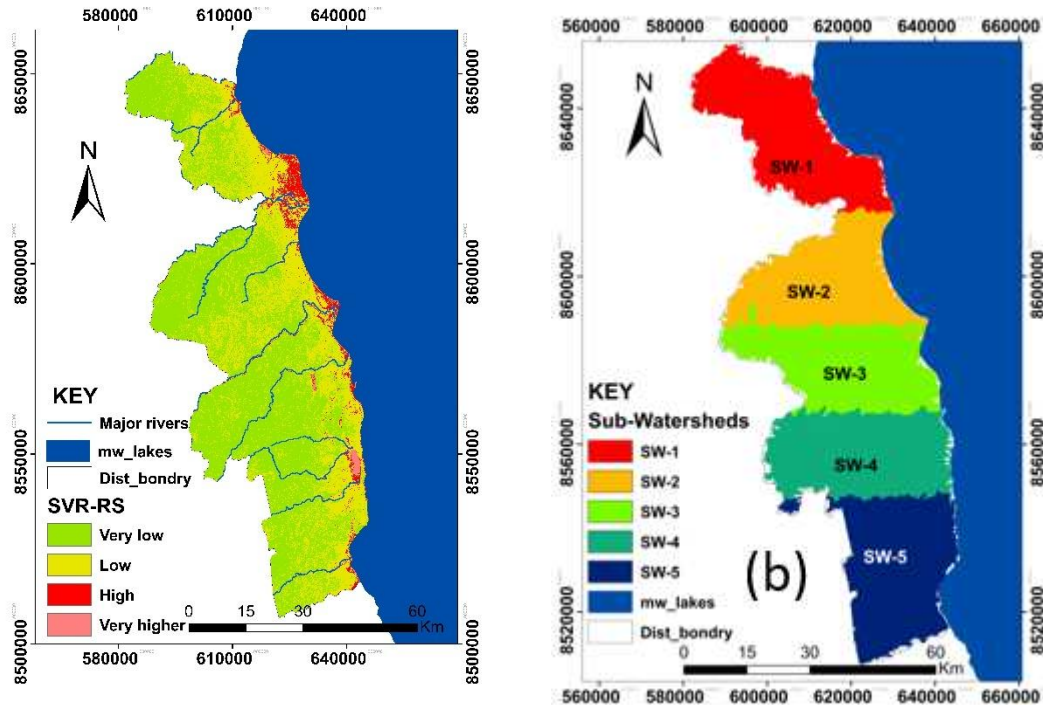


Figure 10. (a) Flood vulnerability map and (b) sub-watersheds of Nkhotakota District.

5. Discussion

5.2 Flood Vulnerability Zoning Analysis

Flood vulnerability zoning in this study was derived primarily from participatory survey data collected among residents living within the watershed. This bottom-up approach prioritizes local knowledge and lived experiences, enabling a nuanced understanding of how flood impacts are perceived and managed at household and community levels. Although expert-driven assessments by flood managers and policymakers are essential for translating findings into policy and structural interventions, they were beyond the scope of the present data collection and remain an important direction for future research. Community-based vulnerability assessments are increasingly recognized as critical complements to biophysical hazard analyses, particularly in regions characterized by limited data availability and high socio-economic sensitivity to floods (Van *et al.*, 2022; Tayyab *et al.*, 2024). The survey findings indicate substantial flood vulnerability across four interconnected dimensions: economic, infrastructural–

physical, institutional–policy, and social–cultural. Economic vulnerability emerged prominently, particularly among households reliant on agriculture and informal income sources. Respondents reported significant challenges in recovering from flood events, largely due to limited savings, absence of financial safety nets, and repeated exposure to flood-related losses. Such conditions have been widely identified as core drivers of flood vulnerability in rural and peri-urban floodplains (Cai *et al.*, 2019; Sibandze *et al.*, 2025). Housing damage and loss were highlighted as especially critical, as recurrent destruction of dwellings contributes to long-term impoverishment, displacement, and heightened livelihood insecurity.

Infrastructural–physical vulnerability was strongly associated with the lack or poor condition of flood mitigation structures, including riverbank protection, drainage systems, roads, and bridges. These deficiencies exacerbate flood impacts by increasing exposure and restricting mobility during emergency and recovery phases. Similar patterns have been documented in flood-prone catchments where weak infrastructure amplifies both direct damages and indirect socio-economic disruptions (Van *et al.*, 2022; Tayyab *et al.*, 2024). Additionally, inadequate enforcement of land-use regulations, particularly concerning construction near river corridors, further intensified physical vulnerability, underscoring the importance of integrated spatial planning in flood risk reduction. Institutional–policy vulnerability was reflected in perceived shortcomings in disaster preparedness, response, and recovery support. Many respondents viewed disaster management initiatives as insufficiently funded, poorly coordinated, or inconsistently implemented. Weak institutional capacity and ineffective policy enforcement have been widely recognized as major contributors to flood vulnerability, especially in developing regions where governance structures face resource and coordination constraints (Frigerio and De Amicis, 2016; Herslund *et al.*, 2016). Limited communication and collaboration between local authorities and higher administrative levels further reduced the effectiveness of risk awareness campaigns and emergency planning efforts. The social–cultural dimension played a decisive role in shaping overall vulnerability. Indicators related to trust in institutions, access to insurance, social cohesion, and flood risk awareness were frequently rated unfavourably by participants. Dissatisfaction with post-flood insurance mechanisms was particularly pronounced, driven by issues of affordability, accessibility, and delayed compensation. These challenges undermine household recovery and weaken confidence in formal risk-transfer systems, consistent with findings from other vulnerability assessments (Nasiri Khiavi *et al.*, 2025; Oluwadare, Chen and McGrath, 2026). At the same time, strong community networks and women’s participation in economic activities were identified as key strengths, reinforcing evidence that inclusive livelihoods and social capital enhance adaptive capacity and resilience to flood hazards (Debnath *et al.*, 2024).

Principal Component Analysis (PCA) provided additional insight into the relative importance and interrelationships among vulnerability dimensions. Although the economic dimension exhibited the highest absolute vulnerability scores, reflecting its severe and direct impacts on livelihoods, the social–cultural dimension accounted for the largest proportion of explained variance, indicating its dominant influence on overall vulnerability patterns. This indicates that community members perceive social cohesion, trust in institutions, and the quality of institutional relationships as critical leverage points for reducing vulnerability under constrained resources. Such findings are consistent with contemporary vulnerability frameworks emphasizing that governance quality and social capital strongly mediate the effectiveness of economic and infrastructural interventions (Membele, Naidu and Mutanga, 2022).

5.1 Flood Risk Patterns, Vulnerability, and Implications for Climate Resilience

Flood risk in Nkhosakota District is shaped by the interaction of physical hazards and community vulnerability. High flood susceptibility was concentrated in low-lying floodplains adjacent to major rivers draining into Lake Malawi, where gentle slopes, high flow accumulation, and shallow groundwater conditions promote prolonged inundation (Hossain and Meng, 2020; Zhang *et al.*, 2020). The SVR–RS model achieved superior predictive performance (AUC = 0.98), highlighting its ability to capture nonlinear relationships between topography, land use, hydrology, and flood occurrence in data-scarce environments (Imran *et al.*, 2019; Macchione *et al.*, 2019). Vegetation covers moderated susceptibility: dense forests in upland areas reduced flooding through enhanced infiltration, while cultivated and deforested floodplains exhibited elevated risk (Avand *et al.*, 2022a).

Vulnerability was strongly influenced by socio-economic, institutional, and physical factors. The economic dimension recorded the highest absolute vulnerability, reflecting low household incomes and reliance on rain-fed agriculture, whereas the social dimension emerged as the most critical relative priority, emphasizing the role of community cohesion, social capital, and trust in institutions (Membele, Naidu and Mutang, 2022b). Institutional limitations, such as weak early warning dissemination and disaster preparedness, and physical constraints,

including poor infrastructure and inadequate riverbank protection, further amplified flood impacts. Spatial integration of susceptibility and vulnerability revealed zones of compounded risk, where high flood-prone areas overlap with high socio-economic and institutional vulnerability, primarily along major river corridors and densely settled floodplains. In contrast, upland sub-watersheds exhibited lower overall vulnerability, despite moderate hazard exposure, illustrating that flood risk is determined more by who and what is exposed than hazard intensity alone (Kourgialas and Karatzas, 2011).

6. Conclusion and Policy Implications

This study provides an integrated assessment of flood susceptibility and community-based vulnerability in Nkhosakota District, revealing clear spatial and socio-economic patterns that shape flood risk. The results demonstrate that sub-watersheds characterized by higher population densities, concentrated infrastructure, and low-lying terrain are disproportionately vulnerable to flooding. In contrast, upstream areas with steeper slopes, higher elevations, and limited human settlement exhibit lower vulnerability, primarily due to reduced exposure rather than lower hazard potential. These findings confirm that flood risk in Nkhosakota is not solely driven by hydrological processes but is strongly mediated by patterns of settlement, land use, and socio-economic conditions. Participatory vulnerability analysis highlighted significant challenges across four interlinked dimensions: economic, infrastructural–physical, institutional–policy, and social–cultural. Economically, flood impacts are particularly severe for households reliant on rain-fed agriculture and informal livelihoods, where housing loss and repeated asset damage undermine long-term recovery and perpetuate poverty cycles. Infrastructural deficiencies, especially the lack of riverbank protection, inadequate drainage, and poor road connectivity, further intensify flood impacts by increasing exposure and constraining emergency response and post-disaster recovery. Institutional weaknesses, including limited disaster preparedness, insufficient funding, and weak coordination between local and national authorities, were perceived as major barriers to effective flood risk reduction. Socially, low trust in formal institutions, limited access to insurance, and uneven risk awareness continue to constrain adaptive capacity, despite the presence of strong informal social networks and active participation of women in household and community economies. The spatial integration of vulnerability indicators revealed distinct high-priority zones within Nkhosakota, where dense settlements coincide with high flood susceptibility. These areas require immediate and targeted intervention. Conversely, sparsely populated upstream sub-watersheds present opportunities for preventive planning, ecosystem-based flood management, and the protection of natural buffers that can reduce downstream flood peaks.

From a policy perspective, the findings underscore the need for a shift from hazard-focused flood management toward people-centered and risk-informed planning. First, land-use zoning regulations must be strengthened and enforced to prevent further settlement within flood-prone river corridors, while promoting safer relocation and flood-resilient housing designs in high-risk zones. Second, investments in both grey infrastructure (e.g., riverbank stabilization, drainage improvement, and resilient road networks) and nature-based solutions (e.g., wetland restoration and riparian buffer protection) are essential to reduce physical exposure. Third, disaster governance in Nkhosakota would benefit from enhanced institutional coordination, increased budgetary support for local disaster management committees, and systematic integration of community knowledge into district-level planning. Finally, expanding access to affordable insurance, livelihood diversification programs, and community-based early warning and awareness initiatives can substantially strengthen social resilience and recovery capacity. Overall, the integrated flood susceptibility and vulnerability maps produced in this study provide a robust scientific and participatory foundation for prioritizing interventions and allocating limited resources. By aligning spatial evidence with local perceptions, this research supports more equitable, effective, and sustainable flood risk management strategies that are directly relevant to climate adaptation and disaster risk reduction efforts in Malawi's flood-prone districts.

Declarations

Funding

This research did not receive any external funding.

Author Contributions

The author conceptualized the study, performed the data analysis, and drafted the manuscript.

Conflicts of Interest / Competing Interests

The authors declare no conflicts of interest. All previously published work referenced in this study has been appropriately cited.

Ethics Approval and Consent to Participate

All participants were fully informed about the study's objectives, procedures, potential risks and benefits, and their rights prior to enrolment. Participation was entirely voluntary, and written informed consent was obtained from all participants. Consent also included authorization to provide contact details, such as telephone numbers or email addresses, solely for research-related communication purposes.

Consent for Publication

Not applicable.

Data Availability

The datasets generated and/or analysed during the current study are available from the corresponding author upon reasonable request.

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